

An Area-Efficient Noise-Shaping SAR ADC With Parallel-Delayed Sampling

Zhengtao Zhu^{ID}, Wenjie Wang, Fan Luo, Longbin Zhu^{ID}, Zhijun Zhou^{ID}, and Keping Wang^{ID}

Abstract—This brief presents an area-efficient noise-shaping (NS) successive approximation register (SAR) analog-to-digital converter (ADC) employing a parallel-delayed sampling (PDS) technique. PDS samples the residual voltages from multiple ADC conversion cycles to increase the NS effect, without the need for large integration capacitors of the typical cascaded passive integrators. A preamplifier is placed between the sampling capacitors and the integrator to avoid signal attenuation, while further reducing the area of the integrator. PDS and preamplifier introduce two left-half-plane poles to the noise transfer function (NTF) to boost the NS effect, while reducing the impact of the parasitic capacitance to essentially enhance the robustness. A prototype 9-bit NS-SAR ADC is designed in a 130-nm CMOS process. At an oversampling ratio (OSR) of 16, the proposed PDS NS-SAR ADC achieves 80.93-dB peak signal to noise and distortion ratio (SNDR) and provides 4.2 NS/area efficiency factor. It consumes a power of 23.46 μ W over a bandwidth of 19.53 kHz, achieving a Schreier figure of merit (FoM_S) of 170.13 dB.

Index Terms—Analog-to-digital converter (ADC), noise shaping (NS), parallel-delayed sampling (PDS), passive integrator, sampling capacitor, successive approximation register (SAR).

I. INTRODUCTION

The successive approximation register (SAR) analog-to-digital converter (ADC) plays a crucial role in medium resolution and moderate sampling rate applications, due to its simple structure and high energy efficiency. However, for high-precision requirements (such as ENOB > 12 bits), it leads to an exponential increase of the capacitor area [1], [2]. Moreover, the accuracy of traditional SAR ADC is limited by comparator noise [3]. To address these challenges, the noise-shaping (NS) SAR ADC has been developed to merge the benefits of the SAR ADC and $\Sigma\Delta$ ADC [2], [5]. The $\Sigma\Delta$ ADC is often employed in demanding high-precision requirements [6], [7], [8], [9]. To achieve NS effect, NS-SAR ADC incorporates an additional filter to process the residual voltage to balance between precision and convention speed with respect to the $\Sigma\Delta$ ADC [10], [11]. Through NS combined with oversampling, the NS-SAR ADC achieves higher ENOB by reducing quantization noise at the expense of SAR ADC conversion speed [12].

The first NS-SAR ADC uses the operational transconductance amplifier (OTA) as an active integrator to achieve the NS effect, but the closed-loop OTA causes large power consumption and area [13]. To achieve a high-resolution and area-efficiency, dynamic amplifiers are used instead of OTA [14]. However, the gain of dynamic amplifiers is sensitive to PVT variation. Alternatively, a fully passive integrator based on switches and capacitors is reported, which

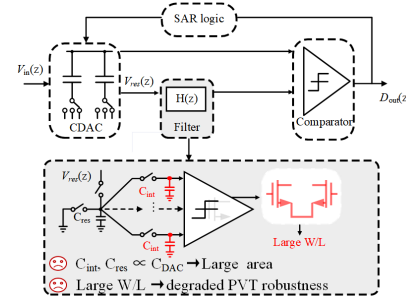


Fig. 1. NS-SAR ADC with cascaded integrators.

provides a simple structure and higher PVT robustness [15]. However, as shown in Fig. 1, the conventional fully passive integrator-based NS-SAR ADC suffers from at least two limitations.

- 1) The typical high-order NS needs cascaded integrators, which need one sampling capacitor (C_{res}) and multiple integration capacitors (C_{int}). These capacitors are proportional to C_{DAC} and occupy a massive area, especially C_{int} (as large as C_{DAC} and much larger than C_{res}).
- 2) The charge transfer between capacitors leads to signal attenuation. Although it can be compensated by the multipath comparator, the cascaded configuration causes an exponential increase in the transistor size of its input pairs and thereby deteriorating the PVT robustness [16], [17], [18], [19], [20].

In this brief, a parallel-delayed sampling (PDS) technique is proposed for NS-SAR ADC. Instead of using multiple C_{int} of cascaded integrators, PDS employs multiple C_{res} for multicycle sampling to enhance the NS effect, which reduces the area as $C_{res} \ll C_{int}$. It also achieves high PVT robustness by avoiding a large transistor ratio of the comparator and suppresses the impact of capacitor mismatch in the passive integrator. Moreover, benefiting from the separation provided by the preamplifier, C_{res} and C_{int} of passive integrators are independent of C_{DAC} , which further reduces the area.

II. PROPOSED ARCHITECTURE

Fig. 2 shows the architecture of the proposed area-efficient NS-SAR ADC. It consists of the PDS, CDAC, preamplifier, passive integrator, comparator, and SAR logic. The CDAC employs a 9-bit capacitor array with bottom-plate sampling to increase the precision. PDS, a three-path preamplifier, a passive integrator, and a two-path comparator form the NS to reduce the quantization noise. To elevate the NS effect at a lower area cost, PDS only consists of sampling capacitor 1 (SC1) and 2 (SC2) to store the residual voltages from multiple conversion cycles. As the black box marked in Fig. 2, SC1 works in every conversion clock cycle. The orange and blue parts of SC2 work alternately in two conversion clock cycles. When either the orange or the blue part of SC2 activates, the other part is disabled. Through this working process, the stored voltages on SC1 and SC2 can be sent to the preamplifier simultaneously. As shown in Fig. 2, the output voltage of the preamplifier is defined as V_{sum} , and the integrating voltage generated by the passive integrator is defined as V_{int} . The schematics of the preamplifier and dynamic comparator are shown in Fig. 3(a) and (b). By adjusting the size ratio of the

Received 14 August 2025; revised 3 November 2025 and 25 November 2025; accepted 7 December 2025. Date of publication 15 December 2025; date of current version 27 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62571114, Grant 62004036, and Grant 62371332; in part by the National Key Research and Development Program of China under Grant 2024YFE0203500; and in part by the Fundamental Research Funds for the Central Universities. (Zhengtao Zhu and Wenjie Wang contributed equally to this work.) (Corresponding author: Zhijun Zhou.)

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Digital Object Identifier 10.1109/TVLSI.2025.3642291

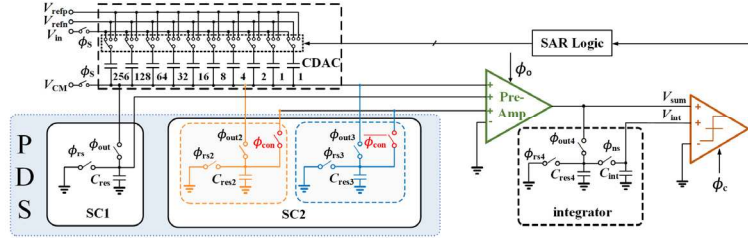


Fig. 2. Architecture of the proposed PDS NS-SAR ADC.

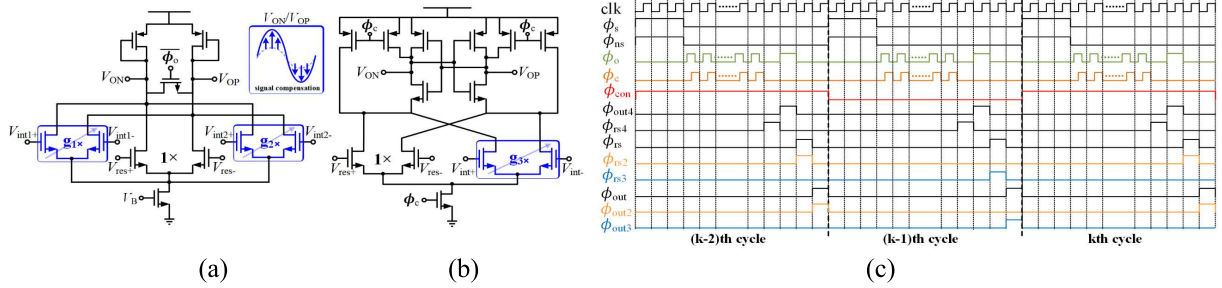


Fig. 3. Schematic of (a) three-path preamplifier, (b) two-path comparator, and (c) timing diagram of PDS NS-SAR ADC.

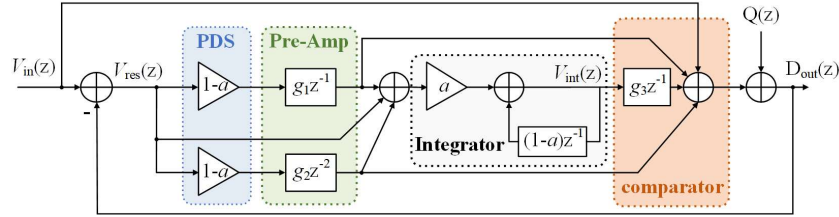


Fig. 4. Signal flow diagram of PDS NS-SAR ADC.

transistors, they determine the coefficients g_1 – g_3 to compensate for signal attenuation on the sampling capacitors and passive integrator.

III. ANALYSIS

A. Basic Operation

Fig. 3(c) shows the timing diagram of the proposed PDS NS-SAR ADC. During the first three clock periods of each ADC conversion cycle, CDAC samples the input voltage as ϕ_s goes high. During the next nine clock pulses, the two-path dynamic comparator sums V_{int} on C_{int} and V_{sum} through ϕ_o and ϕ_c . When the last bit of comparison is completed, a residue voltage V_{res} is generated.

In the following ϕ_{rs4} , the charge on C_{res4} is reset to ground. Then, ϕ_o and ϕ_{out4} go high, the voltages on C_{res} , C_{res2} , or C_{res3} and V_{res} are stored on C_{res4} . During ϕ_{rs} goes high, the voltage on C_{res} is reset. After that, ϕ_{out} goes high, C_{res} samples V_{res} . Then, the sampled voltage on C_{res} will be transferred to C_{res4} in the next cycle after the preamplifier is turned ON. Therefore, the voltages on C_{res} collected by the preamplifier are $V_{res}[k-1]$ in the k th cycle. ϕ_{ns} and ϕ_s are set at the same time. When ϕ_{ns} goes high, charge transfer occurs between C_{res4} and C_{int} to complete the integration.

When ϕ_{out2} goes high in the $(k-2)$ th cycle, C_{res2} samples V_{res} . The voltage on C_{res2} is maintained until ϕ_{con} and ϕ_o are both high, which happens in the k th cycle. When C_{res2} maintains voltage during the $(k-1)$ th cycle, the voltage on C_{res3} is transmitted to C_{res4} and the comparator. Therefore, within one cycle, one of the SC2 holds the voltage, and the other sends the voltage to the preamplifier. Then, it resets and resamples the residual voltage in this cycle, so as to achieve alternating operation of two sampling capacitors. Thus, the voltage on SC2 is $V_{res}[k-2]$ in the k th cycle. $V_{res}[k]$ on the CDAC, $V_{res}[k-1]$ on

C_{res} , and $V_{res}[k-2]$ on C_{res2} (C_{res3}) are simultaneously transferred on C_{res4} to enhance the NS effect. PDS avoids the cascaded integrators by employing a multiple sampling scheme, which not only reduces the area but also evades the large transistor ratio of the comparator to enhance the PVT robustness of the ADC.

B. Signal Flow Diagram

Fig. 4 shows the signal flow diagram of the PDS NS-SAR ADC. Since ϕ_{out2} or ϕ_{out3} is always high when ϕ_{out} is high, the sum of C_{res} and C_{res2} or C_{res3} is assumed to be $a/(1-a)C_{DAC}$. To ensure the voltages on the capacitors are equal, we set $C_{DAC} = C$, $C_{res} = C_{res2} = C_{res3} = aC/2(1-a)$, $C_{res4} = a/(1-a)C_{int}$, and $C_{int} = C/2$. Under such circumstance, $V_{Cres} = V_{Cres2}(V_{Cres3}) = (1-a)V_{res}$. To solve the gain loss during sampling and integration, we configure g_1 – g_3 from the preamplifier and comparator to compensate for the signal attenuation. As mentioned in Section III-A, SC1 samples the residual voltage of the previous clock cycle, and SC2 samples the residual voltage two ADC conversion cycles ago. The voltage on C_{res4} in the z -domain can be expressed as follows:

$$V_{Cres4}(z) = V_{res}(z) [1 + (1-a)g_1z^{-1} + (1-a)g_2z^{-2}]. \quad (1)$$

After being processed by the integrator, the voltage $V_{int}(z)$ on C_{int} can be written as follows:

$$V_{int}(z) = \frac{a}{1 - (1-a)z^{-1}} V_{Cres4}(z). \quad (2)$$

The output $D_{out}(z)$ can be expressed as follows:

$$D_{out}(z) = V_{in}(z) + V_{int}(z)g_3z^{-1} + V_{res}(z)$$

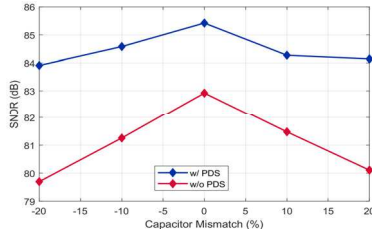


Fig. 5. Simulated SNDR versus capacitor variation.

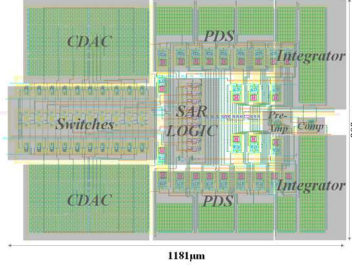


Fig. 6. Layout of the proposed NS-SAR ADC.

$$[(1-a)g_1z^{-1} + (1-a)g_2z^{-2}]. \quad (3)$$

By setting $g_3 = (1-a)/a$, the z -domain transfer function of the entire ADC becomes

$$D_{out}(z) = V_{in}(z) + \frac{1 - (1-a)z^{-1}}{(1-a)g_2z^{-2} + (1-a)g_1z^{-1} + 1} Q(z). \quad (4)$$

The noise transfer function (NTF) can be obtained as follows:

$$NTF(z) = \frac{1 - (1-a)z^{-1}}{(1-a)g_2z^{-2} + (1-a)g_1z^{-1} + 1}. \quad (5)$$

As shown in (5), the position of the zero is determined by a , and the position of the poles is determined by a , g_1 , and g_2 .

Considering the influence of other noise sources, to leave some margin, the signal to noise and distortion ratio (SNDR) is theoretically set to 90 dB, which means the ENOB should be 15. Since the ADC has 9 bits, an additional 6 bits are needed. The oversampling ratio (OSR) is set to 16 to increase 2-bit resolution. Therefore, NS needs to provide 4-bit resolution, which corresponds to an in-band noise suppression of 16 times. Based on the above considerations, we set $a = 1/5$, $g_1 = 2$, $g_2 = 4/5$, and $g_3 = 4$. Therefore, $C_{res} = C_{res2} = C_{res3} = C_{res4} = C/8$ and $C_{int} = C/2$.

Benefiting from PDS and the preamplifier, the total capacitance of the proposed ADC is $2C_{DAC}$, which represents a 60% reduction compared to a conventional 9-bit NS-SAR ADC using cascaded integrators [10]. Moreover, the transistor ratio of PDS (1:2:4/5 for preamplifier and 1:4 for comparator) is much smaller than that of the cascaded integrators (1:4:16 for comparator), which increases the PVT robustness [10].

C. Capacitor Variation

The parameter a is defined by the ratio between the two capacitors of the integrator, but its actual value is exceedingly affected by the parasitic capacitance, which degrades the NS effect. As shown in (5), PDS introduces g_1 and g_2 to the poles, so as to enhance the NS effect and suppress the influence of parasitic capacitance. To investigate the impact of the capacitor mismatch on the NS effect, Fig. 5 shows the SNDR against the capacitor variation of ADCs with and without PDS (disable SC2). A 3-dB degradation occurs when the capacitor mismatch reaches 20%, and it reduces to 1.5 dB by the proposed PDS.

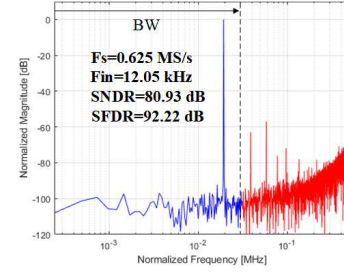


Fig. 7. Postsimulated output spectrum.

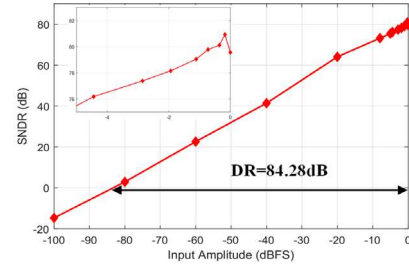


Fig. 8. Postsimulated SNDR versus input amplitude.

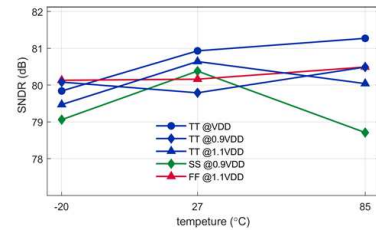


Fig. 9. Postsimulated SNDR under different corners.

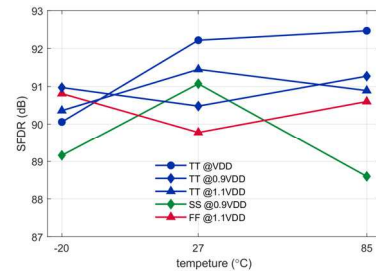


Fig. 10. Postsimulated SFDR under different corners.

IV. RESULTS

The proposed PDS NS-SAR ADC is designed in 130-nm CMOS technology. As shown in Fig. 6, the layout of the ADC core includes differential CDAC arrays, SAR logic, PDS, passive integrators, preamplifier, and comparator. The CDAC is a binary-weighted capacitor array, and it has a total capacitance of 12.64 pF. C_{res} , C_{res2} , C_{res3} , and C_{res4} are 1.58 pF, and C_{int} is 6.32 pF. At the sampling rate of 625 kS/s, the overall power consumption is 23.46 μ W at a 1.2-V supply. The analog and digital circuits consume 16.88 and 6.58 μ W, respectively.

Fig. 7 shows the postsimulated output spectrum (transient noise simulation is enabled) with a 12-kHz input. It achieves an SNDR of 80.93 dB, and the SFDR is 92.22 dB. Fig. 8 shows the postsimulated SNDR versus input amplitude, and it achieves a dynamic range (DR) of 84.28 dB. Figs. 9 and 10 show the postsimulation SNDR and SFDR in 15 different PVT corners. The variations in SNDR and SFDR are

TABLE I
PERFORMANCE SUMMARY AND COMPARISON

Specifications	JSSC '12 [13]	JSSC '19 [17]	TVLSI '24 [18]	TVLSI '25 [19]	TCAS-I '25 [20]	This Work
Technology (nm)	65	40	130	65	65	130
Supply (V)	1.2	1.1	1.2	1.2	1	1.2
Resolution (Bit)	8	9	7	5	10	9
Fs (MS/s)	90	8.4	0.042	600	10	0.625
OSR	4	16	16	15	8	16
BW (kHz)	11000	262	1.3125	20000	625	19.53
SNDR (dB)	62.14	78.4	65.84	70.8	77.7	80.93
ENOB (bit)	10	12.7	10.68	11.5	12.7	13.2
SFDR (dB)	72.46	-	79.6	83.5	88.12	92.22
Power (μ W)	806	143	61	2300	108.3	23.46
FoMs (dB)	163.3	171	-	170.2	175.3	170.13
NS/area efficiency factor ^a	-	1.6	1	-	2.7	4.2

^a Calculated as (ENOB-Original Resolution) / (Total Capacitor- C_{DAC})

limited to 3 and 4 dB, respectively, thus demonstrating substantial robustness.

Table I shows the performance comparison of the proposed ADC with other advanced designs. It can be seen that PDS improves the NS effect by introducing two left-half-plane poles, while avoiding the large C_{int} . It achieves a highest NS/area efficiency factor of 4.2. PDS NS-SAR ADC yields a relatively higher SNDR with lower power consumption and reaching 170.13-dB Schreier FoM (FoM_S). To further enhance the FoM_S, the power consumption should be reduced, such as modifying the preamp as floating-inverter amplifier topology [6].

V. CONCLUSION

This brief proposes a PDS NS-SAR ADC to enhance the NS effect with high area efficiency. It offers two advantages, with respect to cascaded integrators. First, PDS utilizes multiple C_{res} instead of C_{int} to substantially reduce the capacitive area up to 60%. Second, PDS and the separation preamplifier avoid large transistor ratio and suppress the impact of the parasitic capacitance, thereby to increase the PVT robustness. Overall, the proposed architecture is simple, area-efficient, and robust, which achieves an NS/area efficiency factor of 4.2.

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